Employee Payroll System

Java Swing Source Code — Comments Removed

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;
import java.io.*;
import java.util.ArrayList;
class Employee
{
    int id;
    String name;
    double salary;
    String department;
    Employee()
    {
        id = 0;
        name = "";
        salary = 0;
        department = "";
    }
    Employee(int id, String name, double salary, String department)
    {
        this.id = id;
        this.name = name;
        this.salary = salary;
        this.department = department;
    }
    Employee(Employee e)
    {
        id = e.id;
        name = e.name;
        salary = e.salary;
        department = e.department;
    }
    double netSalary()
    {
        return salary + 3000 - 1000;
    }
    String getData()
    {
        return id + "," + name + "," + salary + "," + department;
    }
}
public class EmployeePayroll extends JFrame implements ActionListener
{
    JTextField idField, nameField, salaryField;
    JComboBox<String> deptBox;
    JTextArea area;

    JButton addBtn, saveBtn, loadBtn, viewBtn;

    ArrayList<Employee> list = new ArrayList<Employee>();
    EmployeePayroll()
    {
        setTitle("Employee Payroll System");
        setSize(600, 450);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new BorderLayout(10, 10));

        JPanel form = new JPanel(new GridLayout(4, 2, 5, 5));

        form.add(new JLabel("Employee ID:"));
        idField = new JTextField();
        form.add(idField);

        form.add(new JLabel("Employee Name:"));
        nameField = new JTextField();
        form.add(nameField);

        form.add(new JLabel("Basic Salary:"));
        salaryField = new JTextField();
        form.add(salaryField);

        form.add(new JLabel("Department:"));
        String[] departments = {"IT", "HR", "Sales", "Finance"};
        deptBox = new JComboBox<String>(departments);
        form.add(deptBox);

        add(form, BorderLayout.NORTH);
```

```
    JPanel buttons = new JPanel();

    addBtn = new JButton("Add Employee");
    saveBtn = new JButton("Save");
    loadBtn = new JButton("Load");
    viewBtn = new JButton("View All");

    buttons.add(addBtn);
    buttons.add(saveBtn);
    buttons.add(loadBtn);
    buttons.add(viewBtn);

    add(buttons, BorderLayout.CENTER);

    area = new JTextArea(10, 40);
    area.setEditable(false);

    add(new JScrollPane(area), BorderLayout.SOUTH);

    addBtn.addActionListener(this);
    saveBtn.addActionListener(this);
    loadBtn.addActionListener(this);
    viewBtn.addActionListener(this);

    setLocationRelativeTo(null);
    setVisible(true);
}
public void actionPerformed(ActionEvent e)
{
    try
    {
        if (e.getSource() == addBtn)
        {
            int id = Integer.parseInt(idField.getText().trim());
            String name = nameField.getText().trim();
            double salary = Double.parseDouble(salaryField.getText().trim());
            String dept = (String) deptBox.getSelectedItem();

            if (id <= 0 || name.isEmpty() ||
                salary < 0 || Double.isNaN(salary) ||
                Double.isInfinite(salary) ||
                name.contains(","))
            {
                JOptionPane.showMessageDialog(this,
                    "Enter valid employee details. Do not use comma in name.");
                return;
            }
            for (Employee emp : list)
            {
                if (emp.id == id)
                {
                    JOptionPane.showMessageDialog(this,
                        "This Employee ID already exists.");
                    return;
                }
            }

            Employee emp = new Employee(id, name, salary, dept);
            Employee copy = new Employee(emp);
            list.add(copy);

            JOptionPane.showMessageDialog(this,
                "Employee Added Successfully");

            area.setText("Net Salary: Rs. " + copy.netSalary());

            idField.setText("");
            nameField.setText("");
            salaryField.setText("");
        }
        else if (e.getSource() == saveBtn)
        {
            try (BufferedWriter bw = new BufferedWriter(
                new FileWriter("employees.txt")))
            {
                for (Employee emp : list)
                {
                    bw.write(emp.getData());
                    bw.newLine();
                }
            }

            JOptionPane.showMessageDialog(this,
```

```
                "Records Saved Successfully");
        }
        else if (e.getSource() == loadBtn)
        {
            ArrayList<Employee> temp = new ArrayList<Employee>();

            try (BufferedReader br = new BufferedReader(
                new FileReader("employees.txt")))
            {
                String line;

                while ((line = br.readLine()) != null)
                {
                    String[] data = line.split(",", -1);

                    if (data.length != 4)
                    {
                        throw new IOException("Invalid record in file");
                    }

                    Employee emp = new Employee(
                        Integer.parseInt(data[0]),
                        data[1],
                        Double.parseDouble(data[2]),
                        data[3]);

                    temp.add(emp);
                }
            }

            list.clear();
            list.addAll(temp);

            JOptionPane.showMessageDialog(this,
                "Records Loaded Successfully");
        }
        else if (e.getSource() == viewBtn)
        {
            area.setText("");

            if (list.isEmpty())
            {
                area.setText("No employee records found.");
            }

            for (Employee emp : list)
            {
                area.append(
                    "Employee ID: " + emp.id +
                    "\nName: " + emp.name +
                    "\nDepartment: " + emp.department +
                    "\nBasic Salary: Rs. " + emp.salary +
                    "\nNet Salary: Rs. " + emp.netSalary() +
                    "\n-----------------------------\n");
            }
        }
    }
    catch (NumberFormatException ex)
    {
        JOptionPane.showMessageDialog(this,
            "Enter ID as a whole number and salary as a valid number.");
    }
    catch (IOException ex)
    {
        JOptionPane.showMessageDialog(this,
            "File not found or file has an error.");
    }
}
public static void main(String[] args)
{
    new EmployeePayroll();
}
}
```